

Introduction to Machine Learning

Page 1: Understanding the Basics

Machine Learning is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Instead of writing fixed rules for every situation, developers provide data and a learning algorithm. The algorithm discovers useful relationships within the data and creates a model.

Machine Learning is used in spam detection, recommendation systems, fraud detection, medical analysis, search engines, and many other applications.

A machine learning system commonly includes data, features, an algorithm, and a trained model. Data is the information used for learning, while features are the useful properties extracted from that data.

The quality of a model depends heavily on the quality of its data. Incorrect, incomplete, or biased data can lead to unreliable predictions.

Types of Machine Learning

Page 2: Supervised, Unsupervised, and Reinforcement Learning

Machine Learning is commonly divided into supervised learning, unsupervised learning, and reinforcement learning.

In supervised learning, the model learns from labelled data. Each example contains input features and a known target value. Classification predicts categories, while regression predicts continuous values.

In unsupervised learning, the data does not contain predefined target labels. The model discovers hidden patterns or structures. Clustering and dimensionality reduction are common examples.

Reinforcement learning is based on interaction with an environment. An agent performs actions and receives rewards or penalties. It learns a strategy to maximize total reward.

The choice of learning type depends on the available data, the desired output, and the problem being solved.

The Machine Learning Workflow

Page 3: From Data to Prediction

A machine learning project usually begins with problem definition. The developer decides what needs to be predicted or understood.

The next steps are data collection, data exploration, and data cleaning. Missing values, duplicate records, incorrect data types, and unusual values should be examined.

Feature engineering creates or transforms input variables so that they become more useful for the model. Examples include scaling numbers and converting categories into numerical values.

The dataset is often divided into training and testing portions. Training data is used to learn the model, while testing data evaluates performance on unseen examples.

After training, the model is evaluated using suitable metrics. Classification may use accuracy, precision, recall, F1-score, or ROC-AUC. Regression may use MAE, MSE, RMSE, or R-squared.

Important Machine Learning Concepts

Page 4: Generalization, Overfitting, and Evaluation

A good machine learning model should learn general patterns that work on new, unseen data. This ability is called generalization.

Overfitting occurs when a model memorizes the training data, including noise. It may perform well on training data but poorly on testing data.

Underfitting occurs when a model is too simple to learn the important patterns. It performs poorly on both training and testing data.

Overfitting can be reduced by using more data, simplifying the model, applying regularization, selecting useful features, and using cross-validation.

Data leakage occurs when information that should not be available during training is accidentally used. It can produce unrealistically high evaluation scores.

Machine Learning is an iterative process. Developers may repeatedly improve the data, features, algorithm, and evaluation method.